TOPIC 01 · BONDING IN SOLIDS

Bonding in Solids

Ionic, covalent, metallic and secondary bonds, and how each one shapes the properties of a material.

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Central idea: how the atoms of a material are bonded decides its properties; the properties decide what kind of material it is, and that in turn decides how it can be processed and what it can be used for.

The topic covers the three primary bonds (ionic, covalent and metallic), the weak secondary bonds between molecules, and how each type of bond shows up in the properties of a material.

§ 1 Introduction [Slide 2]

Materials science links four ideas: **structure**, **properties**, **processing** and **applications**. They are often drawn as the four corners of a tetrahedron, because each one is connected to the other three: change the processing and the structure changes, change the structure and the properties change, and so on.

At the root of this chain is the **nature of the bonding** between atoms:

$$\text{Nature of bonding} \leftrightarrow \text{properties} \leftrightarrow \text{type of material} \leftrightarrow \text{processing and applications}$$

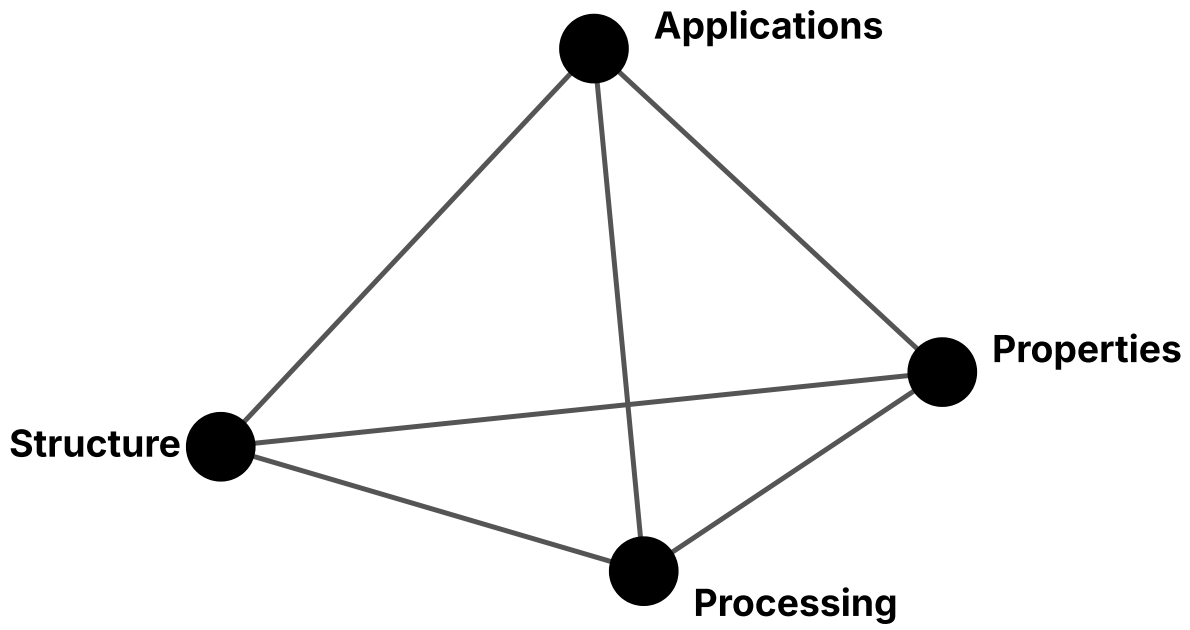


Figure 1. The four linked corners of materials science.

§ 2 Ionic bonding [Slides 4-15]

2.1 What an ionic bond is [Slide 4]

An **ionic bond** forms when one or more valence electrons are **completely transferred** from one atom to another. The result is two ions with opposite charges that attract each other.

- It forms between a **metal** and a **nonmetal**.
- The electron goes from the **less electronegative** atom (the metal) to the **more electronegative** one (the nonmetal). The first becomes a positive **cation**, the second a negative **anion**.
- What holds the ions together is the **electrostatic attraction** between opposite charges.
- This attraction acts equally in all directions: the ionic bond is **non-directional**.

2.2 Example: sodium chloride [Slide 5]

Sodium and chlorine have very different electronegativities. Sodium gives its single valence electron to chlorine; sodium becomes Na^+ and chlorine becomes Cl^-, and the two ions attract each other.

Forming ions also changes their size:

- a metal that becomes a **cation** gets **smaller**;
- a nonmetal that becomes an **anion** gets **larger**.

A perfectly ionic bond is an ideal case. Real ionic bonds always have **some covalent character**.

2.3 How ions are modelled [Slide 6]

To describe ionic solids, ions are treated as:

1. **charged spheres**,
2. **elastic spheres**,
3. **polarizable spheres**: the electron cloud of an ion can be distorted by the ions around it.

The difference with a covalent bond is clear in the electron density: in a covalent bond (such as N_2) the electrons are shared in the region between the two atoms, while in an ionic bond (such as LiF) each ion keeps its own separate electron cloud.

2.4 Force between two ions [Slide 7]

How strongly two ions pull on each other depends on how far apart they are:

- when they are **very far apart**, they do not interact;
- at **large distances**, the Coulomb **attraction** between opposite charges dominates;
- at **short distances**, the electron clouds overlap and **repel**.

The net force is the sum of the two effects:

$$F_{net} = F_{attractive} + F_{repulsive}$$

$$F_{net} = -\frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 a^2} - \frac{nb}{a^{n+1}}$$

Symbol	Meaning
Z_1, Z_2	number of electrons removed or added to each ion (its charge number)
e	electron charge, 1.6×10^{-19} C
ε_0	permittivity of free space, 8.85×10^{-12} $\mathrm{C}^2/(\mathrm{N}\cdot\mathrm{m}^2)$
b	a constant
n	Born exponent, between 5 and 9
a	distance between the ions

If attraction is counted as positive and repulsion as negative, the net force is zero at one particular distance, a_0. Just beyond a_0 the attraction reaches a **maximum**: the largest force the pair can withstand before being pulled apart. When the two ions touch, a_0 equals the cation radius plus the anion radius, $r + R$.

2.5 Energy of an ion pair [Slide 8]

Force and energy describe the same interaction: the force is the derivative of the energy with respect to distance, $F = \partial E / \partial a$. The net energy of the pair is

$$E_{net} = \underbrace{\frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 a_0}}_{\text{attraction: energy released}} + \underbrace{\frac{b}{a_0^n}}_{\text{repulsion: energy absorbed}}$$

Because the two ions have charges of opposite sign, $Z_1 Z_2$ is negative and the attraction term lowers the energy; the repulsion term raises it.

The energy is **lowest** at the **equilibrium distance** a_0, where attraction and repulsion balance. This minimum is the same point where the net force is zero.

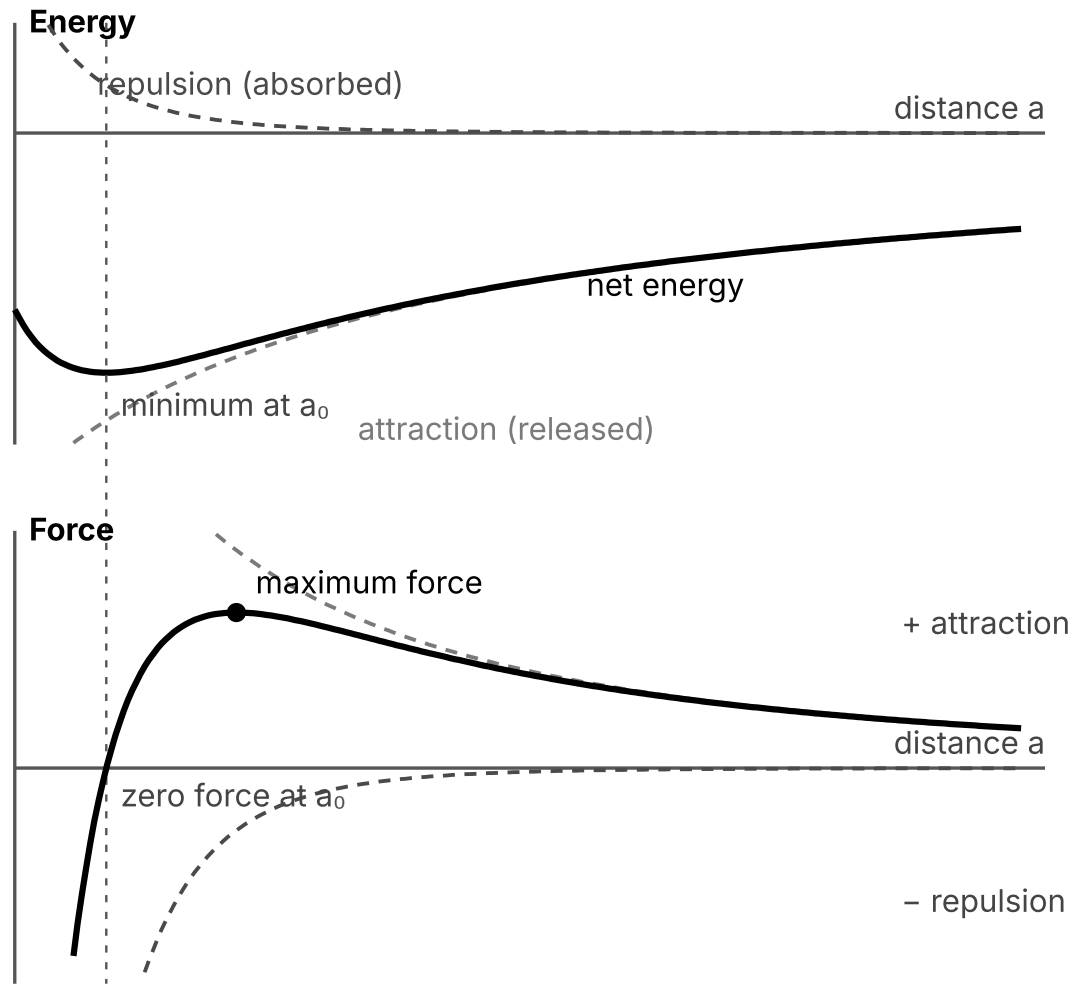


Figure 2. Force and energy between two ions: dashed lines are the attraction (pink) and repulsion (blue) parts, the solid line is their sum. The net energy is lowest at the equilibrium distance a_0, where the net force is zero. Schematic curves, no numerical values.

CHECKING THE LINK BETWEEN FORCE AND ENERGY

Differentiating each term of E_{net} with respect to a gives back the two terms of the force: $\frac{\partial}{\partial a}\left(\frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0}\,a^{-1}\right) = -\frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 a^2}$ and $\frac{\partial}{\partial a}\left(b\,a^{-n}\right) = -\frac{nb}{a^{n+1}}$.

2.6 Lattice energy [Slide 9]

The **lattice energy U** is the energy involved in bringing one mole of ions from infinitely far apart to their fixed positions in a crystal. An equivalent way to see it: the energy **released** when one mole of ionic solid forms from its ions in the gas state.

It follows from the energy of one pair at equilibrium:

$$E_{r=a_0} = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 a_0} + \frac{b}{a_0^n} \quad\Longrightarrow\quad U = -\frac{Z_1 Z_2 e^2 N_A A}{4\pi\varepsilon_0 a_0}\left(1 - \frac{1}{n}\right)$$

where N_A is Avogadro's number (one mole of ion pairs) and A is a constant that depends on the crystal.

WHERE THE FACTOR $(1 - 1/n)$ COMES FROM

1. At a_0 the force is zero: $-\frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 a_0^2} - \frac{nb}{a_0^{n+1}} = 0$.
2. Multiplying by a_0^{n+1} and solving for b: $b = -\frac{Z_1 Z_2 e^2 a_0^{n-1}}{4\pi\varepsilon_0\, n}$.
3. The repulsion term then becomes $\frac{b}{a_0^n} = -\frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 a_0}\cdot\frac{1}{n}$.
4. Adding the attraction term: $E_{r=a_0} = \frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 a_0}\left(1 - \frac{1}{n}\right)$.
5. Multiplying by $N_A A$ gives U. In the expression for U the charge numbers are taken as positive and the minus sign is written in front, so that U is negative.

With n between 5 and 9, repulsion only takes away between 11 % and 20 % of the attraction energy.

The lattice energy cannot be measured directly, but it can be obtained from steps that can be measured, arranged in a closed loop called the **Born-Haber cycle**. For KCl:

Step	ΔH (kJ/mol)
Sublimation of K	+89
Ionization of K(g)	+425
Dissociation of Cl_2(g)	+244
Ionization of Cl(g)	−355
Formation of KCl	−438

CLOSING THE CYCLE

Only half a mole of Cl_2 is needed per mole of KCl, so the dissociation step counts 244/2 = 122 kJ/mol. Going from K(s) + ½Cl_2(g) to the gas ions and then down to KCl(s), the lattice energy is what closes the loop:

$$U = -438 - (89 + 425 + 122 - 355) = -438 - 281 = -719\ \text{kJ/mol}$$

2.7 Properties that depend on the lattice energy [Slides 10-13]

The stronger the bond (the larger the lattice energy), the more energy it takes to separate the ions. Four properties follow from this.

Melting point [Slide 10]

Ionic solid	U (kJ/mol)	U (kcal/mol)	Melting point (°C)
LiCl	829	198	613
NaCl	766	183	801
KCl	686	164	776
RbCl	670	160	715
CsCl	649	155	646
MgO	3932	940	2800
CaO	3583	846	2580
SrO	3311	791	2430
BaO	3127	747	1923

(All the energies are negative; the table gives their size.)

- **Same valence (+1):** larger ions → lower lattice energy → lower melting point.
- **Higher valence (+2):** more charge → much higher lattice energy → much higher melting point. MgO melts at 2800 °C, NaCl at 801 °C.

LiCl does not follow the first trend (it melts lower than NaCl although its U is larger); the trend holds from NaCl to CsCl.

Hardness [Slide 11]

Hardness measures how well a material resists **localized plastic deformation**, caused by pressing into it (indentation) or by scratching (abrasion). Mineralogists use the Mohs scale.

	BeO	MgO	CaO	SrO	BaO
d (Å)	1.65	2.10	2.40	2.57	2.77
Mohs hardness	9.0	6.5	4.5	3.5	3.3

For comparison: hydroxyapatite 5, keratin (fingernail) 2.5.

$$\text{smaller cation } (r_{c^+} \downarrow) \;\Rightarrow\; U \uparrow \Rightarrow \text{ harder}$$

The **binding (potential) energy** is the energy needed to create or break the bond; on the energy-distance curve it is the depth of the minimum. Materials with a high binding energy have **high melting temperatures** and are **hard**.

Thermal expansion [Slide 12]

Thermal expansion is the tendency of matter to change volume when its temperature changes. The **coefficient of thermal expansion** is the change in a linear dimension per degree of temperature, and it is related to the **strength of the bonds**.

When a material is heated, its atoms gain energy and move up inside the energy well:

- in a **deep, narrow** well (strong bonding) the average spacing between atoms barely changes → **low** thermal expansion;
- in a **shallow, wide** well (weak bonding) the spacing grows a lot → **high** thermal expansion.

	NaF	NaCl	NaBr	NaI
α_V (10^{-6} °C^{-1})	39	40	43	48

For comparison: hydroxyapatite $12 \cdot 10^{-6}$ °C^{-1}.

$$\text{larger anion } (r_{a^-} \uparrow) \;\Rightarrow\; U \downarrow \Rightarrow\; \alpha \uparrow$$

Young's modulus [Slide 13]

A high binding energy goes with a **steep** energy curve: a larger force is needed to stretch the bonds, so the **elastic (Young's) modulus is higher**. On a force-distance curve this is the slope dF/da where the curve crosses zero: steeper for strong bonds, gentler for weak ones.

2.8 General properties of ionic compounds [Slide 14]

1. The strong electrostatic attraction gives **high melting and boiling points**.
2. They are **hard and brittle** at room temperature.
3. They are **electrical insulators**, except when molten or dissolved in water. To carry a current, whole ions have to move, and ions move slowly, so the conductivity is poor.
4. They are **soluble in water**.

2.9 How ions are arranged [Slide 15]

- Since the ionic bond is **non-directional**, each ion simply surrounds itself with ions of the opposite charge.
- The arrangement always keeps the solid **electrically neutral** (NaCl and CsCl are typical examples).
- The **smaller the cation compared with the anion**, the **fewer anions** fit around it.
- Ionic compounds are **crystalline solids**.

§ 3 Covalent bonding [Slides 17-21]

3.1 What a covalent bond is [Slide 17]

- It usually forms between **nonmetals**.
- The atoms **share electrons** instead of transferring them.
- It is **highly directional**, and it leaves **no free electrons**.
- It appears between elements with a **small electronegativity difference**, close to each other in the periodic table.
- The outer **s and p electrons** are shared so that each atom reaches the electron configuration of a **noble gas**.

Simple examples: two H atoms share one pair of electrons to form H–H; two Cl atoms share one pair to form Cl–Cl, and each Cl keeps three lone pairs.

3.2 Silicon and silica [Slide 18]

- **Silicon** has a valence of four, so each atom must form **four covalent bonds**. Because these bonds are directional, they point to the corners of a **tetrahedron**, at **109.5°** from each other.
- **Silica (SiO_2)** has the same tetrahedral structure, with covalent bonds between silicon and oxygen.

3.3 Properties of covalent materials [Slide 19]

Covalent solids are built from a **continuous network** of covalent bonds (diamond, boron nitride, quartz SiO_2, silicon carbide SiC). They are:

- **electrical insulators**, both solid and molten;
- **hard and incompressible**;

- of **high melting point** and non-volatile;
- **insoluble**.

Covalent bonds also hold together individual molecules and polymer chains, which are **substances with covalent bonds** rather than covalent networks.

3.4 Polar covalent bonds [Slide 20]

- In a **nonpolar** bond the shared electrons are split **equally** between the two atoms (H–H, Cl–Cl).
- In a **polar** covalent bond the two atoms have **different electronegativities**, so the shared electrons spend more time near one of them (H–Cl). The centres of positive and negative charge no longer coincide.

The size of this separation of charge is the **dipole moment**:

$$\mu = Q \times d$$

with Q the charge and d the distance between the atoms. It is measured in debye: $1\ \mathrm{D} = 3.33 \times 10^{-30}\ \mathrm{C \cdot m}$.

How ionic a bond between atoms A and B is depends on their electronegativities X_A and X_B:

$$\%\ \text{ionic character} = \left(1 - e^{-0.25\,(X_A - X_B)^2}\right) \times 100\%$$

If $X_A = X_B$, the exponential is 1 and the bond is 0 % ionic (purely covalent). As the difference grows, the exponential goes to 0 and the ionic character approaches 100 %, without ever reaching it: a purely ionic bond is an ideal. HI and HBr are nearly covalent, HF is in between, and KCl or LiF are mostly ionic.

3.5 Bond length, bond order and bond energy [Slide 21]

For a given pair of atoms, a **higher bond order** (single → double → triple) makes the bond **shorter**, and a shorter bond is **stronger** (higher bond energy). H_2, F_2 and N_2 follow this rule.

Bond	Bond energy (kJ/mol)	Bond length (nm)
C–C	370	0.154
C=C	680	0.13
C≡C	890	0.12
C–H	435	0.11
O–H	500	0.10
O–Si	375	0.16
F–F	160	0.14
H–H	435	0.074

(The energies are released when the bond forms, so they are negative.)

§ 4 Metallic bonding [Slides 23-27]

4.1 What a metallic bond is [Slide 23]

In a metal, the atoms give up their valence electrons, which form a shared **"sea" of electrons**. What remains of each atom is a positive core, and the cores are held together by their attraction to the negative electron sea. In short: the metallic bond is the **electrostatic attraction between metal cations and delocalized electrons**.

- The atoms are **closely packed** in a crystal structure.
- The electrons are spread among all the atoms as an **electron cloud**.
- Since each outer electron is shared by many atoms, the bond is **non-directional**.

4.2 Bonding energy and valence electrons [Slide 24]

- Forming metallic bonds **lowers the energy** of the atoms.
- As for any bond, the energy is lowest at the **equilibrium distance** a_0.
- **Fewer valence electrons → weaker bond.** Potassium (valence +1): bonding energy 89.6 kJ/mol, melting point 63.5 °C.
- **More valence electrons → stronger bond.** Calcium (valence +2): bonding energy 177 kJ/mol, melting point 851 °C.

Bonding energies and melting points of the fourth-period metals:

Element	Electron configuration	Bonding energy (kJ/mol)	Melting point (°C)
K	$4s^1$	89.6	63.5
Ca	$4s^2$	177	851
Sc	$3d^1 4s^2$	342	1397
Ti	$3d^2 4s^2$	473	1660
V	$3d^3 4s^2$	515	1730
Cr	$3d^5 4s^1$	398	1903
Mn	$3d^5 4s^2$	279	1244
Fe	$3d^6 4s^2$	418	1535
Co	$3d^7 4s^2$	383	1490
Ni	$3d^8 4s^2$	423	1455
Cu	$3d^{10} 4s^1$	339	1083
Zn	$4s^2$	131	419
Ga	$4s^2 4p^1$	272	29.8
Ge	$4s^2 4p^2$	377	960

4.3 What metallic bonding means for properties [Slide 25]

- Pure metals are much more **malleable** than ionic or covalent networks. Their strength can be raised considerably by **alloying**.
- Pure metals are **excellent conductors of heat and electricity**: under a voltage, the electrons of the electron sea move freely and carry the current.

4.4 Band theory [Slides 26-27]

Band theory describes the electrons of a solid, which can only have energies inside certain ranges.

When atoms join into a metallic crystal, the orbitals of the separate atoms combine into a set of orbitals that belong to the whole crystal. Each new atom adds more levels, and with a huge number of atoms the levels are so close together that they form an almost continuous range of energies: a **band**. A piece of metal behaves like a single giant molecule. Lithium shows it: Li, Li_2, Li_3, ..., Li_N, with the 2s levels multiplying into a band.

The band structure depends on the **distance between atoms**: the bands form because the outermost orbitals of neighbouring atoms overlap. In lithium and in magnesium this gives a valence band and a conduction band.

> NOTE ON THE OFFICIAL MATERIAL
> The magnesium example is labelled "Z = 4, $1s^2\ 2s^2$". That is the configuration of beryllium; magnesium has Z = 12.

§ 5 Secondary bonding: intermolecular forces [Slides 29-35]

5.1 Primary bonds are strong [Slide 29]

Primary bond	Energy (kcal/mol)	Mechanism
Ionic	150-370	electrons transferred → cations and anions → Coulomb force (non-directional)
Covalent	125-300	electrons shared (directional)
Metallic	25-200	electrons shared by all the atoms, held only loosely by the nuclei (non-directional)

5.2 Secondary bonds are weak [Slide 30]

Secondary (van der Waals) bonds are attractions between **electric dipoles** in atoms or molecules. A dipole appears when the electron charge is spread **unevenly**, so that the centre of positive charge and the centre of negative charge are in different places.

There are two kinds of dipole:

- **permanent dipoles**, which do not change with time;
- **fluctuating dipoles**, which appear and disappear as the electron cloud changes with time.

5.3 Permanent dipoles (1-10 kcal/mol) [Slide 31]

They form in molecules that:

- have a **constant dipole moment μ**, and
- are made of atoms of **different electronegativity** joined by **covalent** bonds.

Whether a molecule has a dipole depends on its shape:

- **CH_4:** the four C–H bonds are arranged **symmetrically**; their effects cancel and there is **no dipole moment**.
- **CH_3Cl:** replacing one H by Cl makes the tetrahedron **asymmetrical**, and the molecule **has a dipole**.

5.4 Example: PVC [Slide 32]

In polyvinyl chloride (PVC) the chlorine atoms along the chain carry a negative charge and the hydrogen atoms a positive one. Neighbouring chains therefore attract each other through van der Waals bonds, and this extra bonding makes PVC **stiffer**. When a force is applied, these weak bonds break and the chains **slide past one another**.

5.5 The hydrogen bond (≈ 7 kcal/mol in water) [Slide 33]

The **hydrogen bond** is a strong case of permanent dipole. When hydrogen is bonded to a very **electronegative atom** (O–H, N–H, F–H), the charge is distributed very unevenly: the bond is polar and forms a permanent dipole. A hydrogen bond is the **dipole-dipole attraction between polar bonds that contain hydrogen**.

In **water**, the two hydrogens are not in a straight line with the oxygen (the H–O–H angle is about 105°), so the molecule is a dipole. The **oxygen** is the **negative** end and the **hydrogens** the **positive** end, and the hydrogen of one molecule is attracted to the oxygen of the next.

> NOTE ON THE OFFICIAL MATERIAL
> The official text writes "positive oxygen pole and negative hydrogen pole". The signs are swapped: oxygen, being more electronegative, pulls the shared electrons and is the negative pole.

5.6 Fluctuating dipoles (< 2 kcal/mol) [Slide 34]

Even **nonpolar** molecules attract each other weakly. At any instant the electron cloud may be slightly lopsided, creating a short-lived dipole; a moment later it is symmetrical again or lopsided the other way. These **fluctuating dipoles** are the weakest of the secondary bonds.

5.7 Example: boiling points of the hydrides [Slide 35]

Going down a group (to higher periods), the boiling points of the hydrides generally rise, because larger atoms with more electrons give stronger forces between the molecules (CH_4 → SiH_4 → GeH_4 → SnH_4).

Water, HF and NH_3 break this trend: their **hydrogen bonds** are much stronger than ordinary van der Waals forces, and so their boiling points are **far higher** than the trend would predict.

§ 6 Bonding and properties in materials [Slides 37-38]

The type of bond largely defines the family of material:

Type of material	Character of bonding	Examples
Metals	Metallic	Fe, steels
Ceramics and glasses	Ionic / covalent	Silica (SiO_2)
Polymers	Covalent and secondary	Polyethylene $-(CH_2)-$

Summary of properties by type of bond:

Bond	Type of substance	Melting and boiling points	Mechanical properties	Solubility
Covalent	Molecular	Low	Soft in the solid state	Depends on the polarity of the molecules
Covalent	Atomic network (lattice)	Very high	Very hard, brittle	Insoluble in all solvents
Ionic	Ionic	High	Hard and brittle	Soluble in polar solvents
Metallic	Metallic	High	Ductile and workable	Insoluble in all solvents

§ 7 Questions [Slide 39]

1. Compare metallic and covalent primary bonds: nature of the bond, valence of the atoms involved, and ductility of the materials
2. Titanium is stiffer than aluminium, has a lower thermal expansion coefficient and a higher melting temperature. Sketch both potential-well curves on the same graph and show how these properties appear in them
3. Would MgO or magnesium have the higher modulus of elasticity?
4. Aluminium and silicon are neighbours in the periodic table. Which would have the higher modulus of elasticity?

Question 1

Metallic vs Covalent Bonding

Compare and contrast metallic and covalent primary bonds in terms of (a) the nature of the bond, (b) the valence of the atoms involved, and (c) the ductility of the materials bonded in these ways.

Answer

(a) Nature of the bond. A covalent bond is a pair of electrons **shared between two specific atoms**, so it points in fixed directions (e.g. 109.5° in silicon). A metallic bond is the attraction between positive cores and a **sea of delocalized electrons** shared by all the atoms, so it has no direction.

(b) Valence. In covalent bonding the valence fixes **how many bonds** an atom must form: silicon, with valence 4, forms exactly four. In metallic bonding the valence fixes **how many electrons go to the sea**: the more valence electrons, the higher the bonding energy and the melting point (K → Ca).

(c) Ductility. Covalent materials are hard and brittle: their bonds have fixed directions. Metals are ductile, malleable and workable: their bonds are non-directional because the electron sea is shared by all the atoms.

Interpretation

The two differences, **directional vs non-directional** and **localized vs delocalized** electrons, explain the rest: the same pair of ideas gives conduction in metals (the electron sea moves under a voltage) and its absence in covalent solids (no free electrons, non-conductive lattices).

Question 2

Potential Curves of Titanium vs Aluminium

Titanium is stiffer than aluminum, has a lower thermal expansion coefficient than aluminum, and has a higher melting temperature than aluminum. On the same graph, carefully and schematically draw the potential well curves for both metals and showing how the physical properties are manifested in these curves.

Answer

Draw both curves on the same axes (energy vs distance), with their minima close to each other in distance:

- **Titanium:** a **deeper** minimum (higher melting temperature), a **narrower** trough (lower thermal expansion) and **steeper** walls around a_0 (stiffer).
- **Aluminium:** a **shallower** minimum, a **wider** trough and gentler walls.

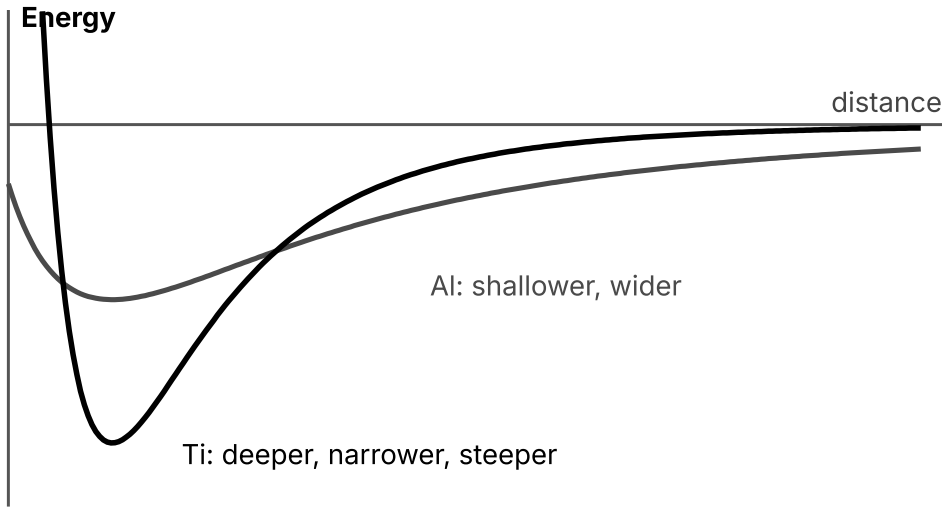


Figure 3. Schematic potential wells: the deeper Ti minimum gives the higher melting temperature, the narrower trough the lower thermal expansion, the steeper walls the higher stiffness. Shapes only, no numerical values.

How each property shows up:

1. **Melting temperature:** the Ti minimum lies lower; more energy is needed to break its bonds.
2. **Thermal expansion:** heating adds the same energy to both curves; in the narrow Ti trough the spacing between atoms grows less than in the wide Al trough.
3. **Stiffness:** around a_0 the Ti curve is steeper, so a larger force is needed to stretch the bonds.

Interpretation

All three facts point to the same thing: the bonds in titanium are **stronger** than in aluminium. One curve shape, deeper, narrower and steeper, carries the three properties at once.

Question 3

Elastic Modulus of MgO vs Mg

Would you expect MgO or magnesium to have the higher modulus of elasticity? Explain.

Answer

MgO should have the higher modulus of elasticity. Its ionic bond between Mg^{2+} and O^{2-} has a very high binding (lattice) energy, so its energy curve is deep and steep and a greater force is needed to stretch the bonds. Metallic Mg has a weaker, non-directional bond to an electron sea, which gives a softer, more malleable material.

Interpretation

The same strong ionic bonding that gives MgO its high melting point (2800 °C) and hardness also gives it its stiffness: all of them come from the deep, steep energy curve of a high-binding-energy bond.

Question 4

Elastic Modulus of Aluminium vs Silicon

Aluminum and silicon are side-by-side in the periodic table. Which would you expect to have the higher modulus of elasticity (E)? Explain.

Answer

Silicon should have the higher modulus of elasticity. It is held together by a continuous network of strong, **directional** covalent bonds, which makes it hard and incompressible: a greater force is needed to stretch the bonds, which is what a high modulus means. Aluminium is held by **non-directional** metallic bonds to an electron sea, which gives a malleable, workable material.

Interpretation

Two elements next to each other in the periodic table can behave very differently because they bond differently: covalent networks are very hard and brittle, metals are ductile and workable. Silicon's higher stiffness is the mechanical side of that difference.